



NCCER Module 22107



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Vertical-Mast Sit-Down Counterbalance Forklifts





Section Four

Forklift Attachments

Objective

4. Describe the types of attachments and how they extend a forklift's capabilities.
 - a. Describe how to determine what attachment should be used.
 - b. Explain why the use of any attachment introduces new hazards and safety risks.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

3. Interpret forklift load charts used with attachments.



4.0.0 Forklift Attachments

Specialized attachments can be used on conventional forklifts to perform a variety of different jobs.



4.1.0 Selecting the Correct Attachment

To select the necessary attachment, consider the following:

- ***Will you be lifting odd-shaped loads?*** Loads that are long or asymmetrical may require the use of specialized forks, a boom, lifting clamp attachments, etc.
- ***Will you need to reach out to lift the load?*** The vertical mast on a conventional forklift does not extend like a boom-type forklift. Ensure that the attachment allows you to perform this function if necessary.
- ***Will the job require you to lift personnel and equipment?*** The use of **personnel platforms** brings extra risks to workers, so it is critical to understand precautions that need to be taken in these situations.



4.2.0 Attachment Types

The most common attachments for vertical mast forklifts are:

- **Specialized forks**
- Paper and **barrel clamps**
- **Flat surface clamps**
- **Extension booms** and hooks
- Personnel platforms
- **Floor brooms**
- **Push-Pulls**



4.2.1 Specialized Forks

- Specialized forks up to 96" long and 7" wide are available for lifting lumber and other large non-symmetrical items.
- Some models have forks that can be adjusted wider.



4.2.2 Barrel Clamps

- Barrel clamps use two hydraulic arms to clamp a roll or barrel and safely transport it to the desired location.
- Many barrel clamp attachments can handle multiple rolls or barrels at a time, depending on the load capability of the forklift.



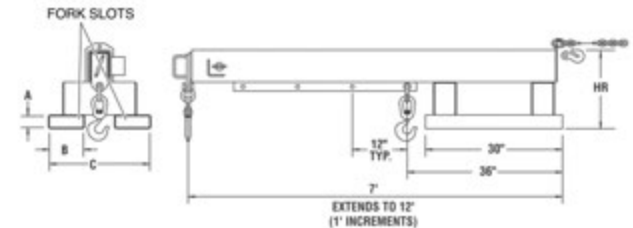
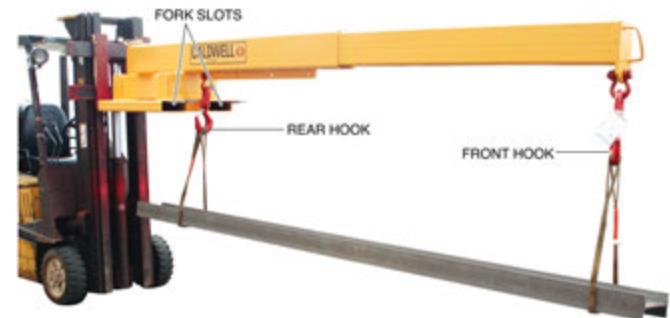
4.2.3 Flat Surface Clamps

- Appliances are often loaded and shipped without pallets. In these cases, using standard forks to load or unload them can damage the box or appliance.
- Flat surface clamps work well in these situations because they grasp the load by the sides.



4.2.4 Extension Booms and Hooks (1 of 2)

- Extension booms and lifting hooks allow the conventional forklift to work like a small crane with loads that must be lifted from the top.
- Boom extensions can be either fixed angle or adjustable angle.
- The operator must understand how to use the **load chart**.
- The machine's lifting capacity changes based on its distance and angle from the load.



SPECIFICATIONS

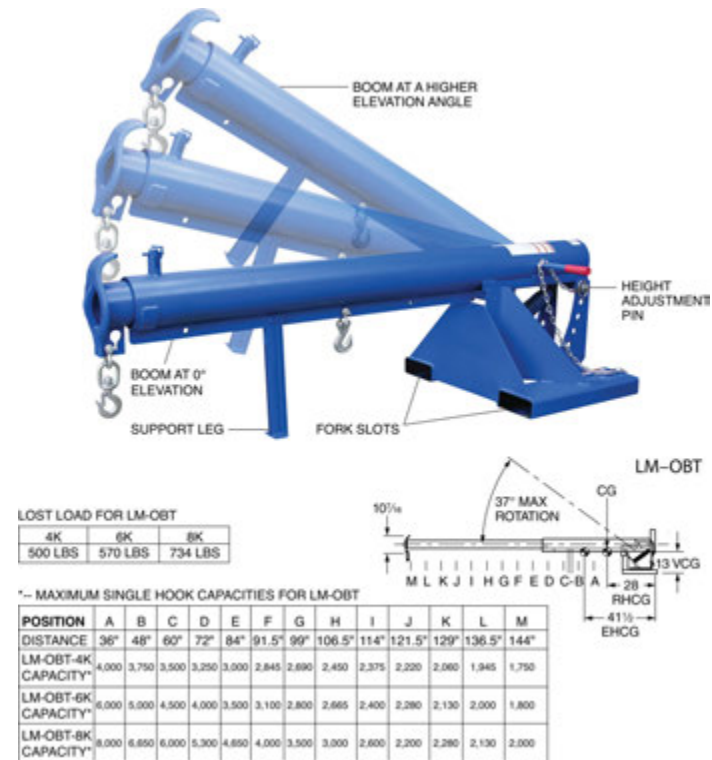
Model Number	Dimensions (in.)				Maximum Capacity @ Hook Position (lbs.)								Weight (lbs.)
	A	B	C	HR	Ø	3'-6"	7'	8'	9'	10'	11'	12'	
FB-30	2½	7½	22	16	1.00	3,000	3,000	2,600	2,200	1,900	1,600	1,500	340
FB-40	16	1.09	4,000	3,200	2,600	2,200	1,900	1,600	1,500	340			
FB-60	17	1.36	6,000	5,000	4,200	3,500	3,000	2,700	2,500	390			
FB-80	18	1.51	8,000	7,000	5,700	4,800	4,100	3,600	3,100	390			

WARNING

Capacities are for boom only and are dependent on fork lift used. Check with fork lift manufacturer for load capacities before use.

4.2.4 Extension Booms and Hooks (2 of 2)

- Adjustable-angle booms are excellent for loads lifted on varying elevation angles.
- The weight of the extension boom must be considered when determining the total weight that the forklift can handle.
- The closer the load is to the boom's base, the more load the boom can lift.



4.2.5 Personnel Platform

- Personnel platforms must adhere to OSHA 1926.451.
- These platforms can only be used on forklifts designed to support them.
- Personnel must use personal fall protection equipment (PPE).
- The platform must be securely attached to the forks before lifting personnel.
- Forklift operators must never travel with workers on the platform.
- Overhead clearance must be confirmed before lifting workers.
- Forklift operators must remain seated at the controls of the forklift while workers are on the platform.



4.2.6 Floor Brooms and Other Attachments

- The floor broom, or powered broom, is a large broom head that can be attached to the forks.
- Operators can use broom attachments to clean parking lots, industrial plants, warehouses, mill yards, or any large area where the floor or ground is smooth and level.
- Other types of attachments include slip sheets, bale clamps, carton clamps, and side shifters.



Wrap Up – Trade Terms (1 of 2)

Barrel Clamps: Attachments that use two hydraulic arms to clamp a roll or barrel and safely transport it to the desired location.

Extension Booms: Attachments that allow the conventional forklift to work like a small crane.

Flat Surface Clamps: Attachments used to move loads that are flat sided or rectangular.

Floor Brooms: Large broom heads that can be attached to the forks to clean parking lots, industrial plants, warehouses, mill yards, or any large area where the floor or ground is smooth and level.



Wrap Up – Trade Terms (2 of 2)

Load Chart: Used by an operator, when using a boom attachment, to determine the forklift's lifting capacity based on a load's distance and angle from the forklift.

Personnel Platforms: Attachments that provide an elevated work station that can hold workers, along with their tools and other items.

Push-Pulls: Attachments that allow forklifts to work with loads stacked on slip sheets, which are flat plastic sheets that replace pallets. These are often used in bottling or canned goods handling.

Specialized Forks: Forks up to 96" long and 7" wide that are available for lifting lumber and other large items.



4.0.0 Section Review Question 1

1. Which of the following is a forklift attachment?
 - a. Carriage
 - b. Extension boom
 - c. Load backrest
 - d. Counterweight



4.0.0 Section Review Question 1 Answer

1. Which of the following is a forklift attachment?
 - a. Carriage
 - b. Extension boom**
 - c. Load backrest
 - d. Counterweight



4.0.0 Section Review Question 2

2. What load factors should be considered when determining whether a forklift attachment is needed?
- a. Cost of the load
 - b. Shape of the load
 - c. Height the load will be lifted
 - d. How the load will be shipped



4.0.0 Section Review Question 2 Answer

2. What load factors should be considered when determining whether a forklift attachment is needed?
- a. Cost of the load
 - b. Shape of the load**
 - c. Height the load will be lifted
 - d. How the load will be shipped



4.0.0 Section Review Question 3

3. When an attachment is placed on a forklift, what changes?
- a. Load capacity
 - b. Lifting capacity
 - c. Lifting height
 - d. Center of gravity



4.0.0 Section Review Question 3 Answer

3. When an attachment is placed on a forklift, what changes?
- a. Load capacity
 - b. Lifting capacity**
 - c. Lifting height
 - d. Center of gravity



4.0.0 Section Review Question 4

4. When placing an attachment on a forklift, what is important to consider when determining the total weight that the forklift can handle?
- a. Forklift class
 - b. Attachment weight
 - c. Fork weight
 - d. Mast type



4.0.0 Section Review Question 4 Answer

4. When placing an attachment on a forklift, what is important to consider when determining the total weight that the forklift can handle?
- a. Forklift class
 - b. Attachment weight**
 - c. Fork weight
 - d. Mast type



Next...

Section 5.0.0

Basic Preventive Maintenance



Read Sections 5.0.0 through 5.2.2 and complete the Section Review.